FI 362: Linear Time Series Analysis and Its Applications

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Based on Tsay, *Analysis of Financial Time Series*, Ch. 2.

Where we are, and why this chapter matters

- **Chapter 1** asked: *what does a return look like?*
  - (distribution, fat tails, non-normality)
- **Chapter 2** asks the next question: *Is today's return predictable from its own past? And how do we build, check, and forecast a model that captures any such dependence?*

Three questions we will answer with real data

1. Why are *prices* non-stationary but *returns* stationary?
2. Are S&P 500 returns serially correlated?
3. Why does volatility have "long memory" when returns do not?

Roadmap

Stationarity

Stationarity: the foundation of all inference

- To learn anything from one realisation of $\{r_t\}$, its statistical properties must not drift over time.

Strict stationarity

The joint distribution of $(r_{t_1}, \ldots, r_{t_k})$ equals that of $(r_{t_1+\tau}, \ldots, r_{t_k+\tau})$ for *all* τ, k.

Very strong; essentially impossible to verify empirically.

Weak (covariance) stationarity

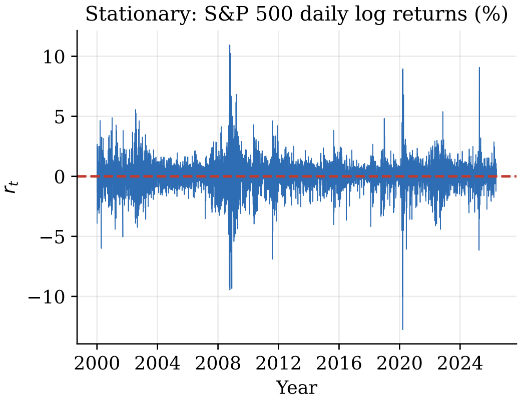
1. $\mathbb{E}(r_t) = \mu$ (constant mean)
2. $\mathrm{Cov}(r_t, r_{t-\ell}) = \gamma_\ell$ depends *only* on the lag ℓ

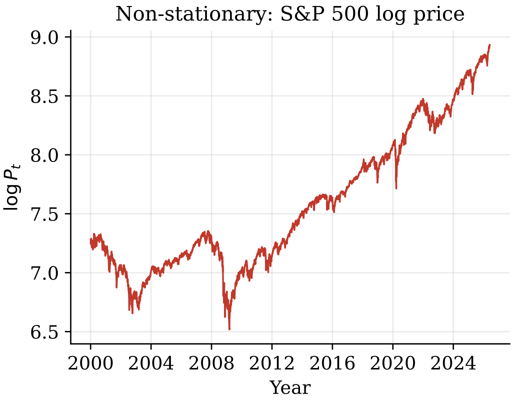
Tractable; the working assumption in finance.

Read it off a time plot

A weakly stationary series fluctuates around a *fixed level* with *constant spread.* If the level wanders or the spread grows, suspect non-stationarity. (Under normality: weak $\equiv$ strict.)

Stationary vs. non-stationary: see the difference





Left: S&P 500 daily *log returns* – fixed level near 0, roughly constant spread (bursts aside). *Stationary.*

Right: S&P 500 *log price* – a wandering level with no tendency to return. *Non-stationary* (a unit root; §2.7).

Same asset, one difference apart – this gap drives the whole chapter.

Correlation and the Autocorrelation Function

Correlation

- The correlation coefficient between two random variables X and Y is defined as

$$\rho_{x,y} = \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}} = \frac{E[(X-\mu_x)(Y-\mu_u)]}{\sqrt{E(X-\mu_x)^2 E(Y-\mu_y)^2}}$$

  where,
  - μ_x is the mean of X and μ_y is the mean of Y, assuming variances exist
  - $-1 \leq \rho_{x,y} \leq 1$ and $\rho_{x,y} = 0$ if and only if (iff) X & Y are independent
- Correlation for given sample of $\{(x,y)\}_{t=1}^{T}$ can be calculated

$$\hat{\rho}_{x,y} = \frac{\sum_{t=1}^{T}(x_t-\bar{x})(y_t-\bar{y})}{\sqrt{\sum_{t=1}^{T}(x_t-\bar{x})^2 \sum_{t=1}^{T}(y_t-\bar{y})^2}}$$

The autocorrelation function (ACF)

- Linear dependence between r_t and its past values r_{t-i}, correlation $\to$ **autocorrelation**
- For a weakly stationary $\{r_t\}$, the **lag-ℓ autocorrelation** is

$$\rho_\ell = \frac{\mathrm{Cov}(r_t, r_{t-\ell})}{\sqrt{\mathrm{Var}(r_t)\,\mathrm{Var}(r_{t-\ell})}} = \frac{\gamma_\ell}{\gamma_0}, \qquad \text{where} \quad \mathrm{Var}(r_t) = \mathrm{Var}(r_{t-\ell})$$

- **Sample ACF** (size T):

$$\hat{\rho}_\ell = \frac{\sum_{t=\ell+1}^{T}(r_t - \bar{r})(r_{t-\ell} - \bar{r})}{\sum_{t=1}^{T}(r_t - \bar{r})^2}$$

- $\{r_t\}$ serially uncorrelated $\iff \rho_\ell = 0\ \forall \ell > 0$
- $\rho_0 = 1,\ \ \rho_\ell = \rho_{-\ell},\ \ |\rho_\ell| \leq 1$

Testing for serial correlation

- If $\{r_t\}$ is i.i.d. with finite variance, $\hat{\rho}_\ell \xrightarrow{d} \mathcal{N}(0, 1/T)$.
- **Single lag** $(H_0 : \rho_\ell = 0)$

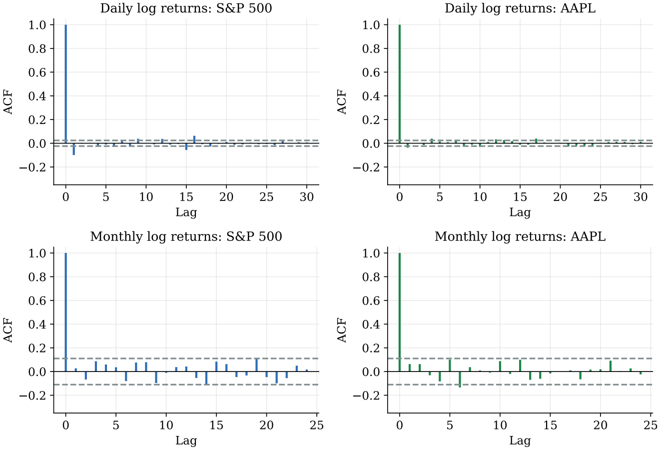
$$t = \frac{\hat{\rho}_\ell}{\sqrt{(1 + 2\sum_{i=1}^{\ell-1} \hat{\rho}_i^2)/T}} \sim \mathcal{N}(0,1)$$

  Reject H_0 if $|t| > Z_{\alpha/2}$ where $Z_{\alpha/2}$ is the $100(1-\alpha/2)$th percentile of standard normal distribution.
- Many software packages use $1/T$ as the asymptotic variance of $\hat{\rho}_\ell$ $\forall \ell \neq 0$.
- **Ljung–Box** $(H_0 : \rho_1 = \cdots = \rho_m = 0)$

$$Q(m) = T(T+2)\sum_{\ell=1}^{m} \frac{\hat{\rho}_\ell^2}{T-\ell} \xrightarrow{d} \chi_m^2$$

  Reject if $p < \alpha$. Rule of thumb: $m \approx \ln T$.

Testing for serial correlation



- The dashed bands on every ACF plot are $\pm 1.96/\sqrt{T}$: spikes inside are “statistically zero”.

White Noise and Linear Time Series

White noise & Linear Time Series

White noise

A time series r_t is **white noise** if $\{r_t\}$ is a sequence of i.i.d., with mean zero, $\mathbb{E}(a_t) = 0$, and finite variance, meaning all ACFs zero.
Gaussian WN if $r_t \sim \mathcal{N}(0, \sigma_a^2)$

Linear Time Series

A time series r_t is said to be linear if it can be written as

$$r_t = \mu + \sum_{i=0}^{\infty} \psi_i \, a_{t-i}, \quad \psi_0 = 1$$

where ψ-weights are the **impulse response**: the echo of a shock today and $\{a_t\}$ is a white noise series.

- a_t denotes the *innovation* or *shock* at time t, and assumed continous random variable in this lecture series even though not all financial time series are linear.

Linear time series

$$r_t = \mu + \sum_{i=0}^{\infty} \psi_i \, a_{t-i}, \quad \psi_0 = 1$$

- Dynamic structure of r_t is governed by coefficients of ψ_i
- If r_t is weakly stationary, its mean and variance can be obtained by using independence of $\{a_t\}$ as

$$E(r_t) = \mu \qquad \mathrm{Var}(r_t) = \sigma_a^2 \sum_{i=0}^{\infty} \psi_i^2$$

- If $\mathrm{Var}(r_t) < \infty$, $\{\psi_i^2\}$ must be a convergent sequence, i.e., $\psi_i^2 \to 0$ as $i \to \infty$.
- Covariance coefficient is

$$\rho_\ell = \frac{\sum_{i=0}^{\infty} \psi_i \psi_{i+\ell}}{1 + \sum_{i=0}^{\infty} \psi_i^2}$$

Autoregressive (AR) Models

Why an AR model? Start from what the data told us

- In the previous section we found that the **S&P 500 value-weighted index** has a *statistically significant* lag-1 autocorrelation ($\hat{\rho}_1 \approx -0.10$, well outside the $\pm 2/\sqrt{T}$ band).
- Read plainly: *yesterday's return r_{t-1} is correlated with today's return r_t* – so r_{t-1} may carry information that is useful for *predicting* r_t.

The natural next step

If r_{t-1} helps predict r_t, then *use it as a regressor*: model today's return as a straight-line function of yesterday's return plus a fresh, unpredictable shock. A regression of a series on *its own past* is exactly an **autoregressive (AR) model**.

One significant lag ⇒ a one-line model

$$r_t = \phi_0 + \phi_1 r_{t-1} + a_t \qquad \text{(the AR(1)).}$$

AR(1): the simplest memory

$$r_t = \phi_0 + \phi_1 r_{t-1} + a_t, \qquad \{a_t\} \sim \mathrm{WN}(0, \sigma_a^2)$$

In words: today's return = a constant + a fraction ϕ_1 of yesterday's return + a fresh shock a_t that cannot be predicted from the past.

Properties when $|\phi_1| < 1$ (weakly stationary)

$$\mathbb{E}(r_t) = \frac{\phi_0}{1 - \phi_1}, \qquad \mathrm{Var}(r_t) = \frac{\sigma_a^2}{1 - \phi_1^2}, \qquad \rho_\ell = \phi_1^\ell.$$

Take expectations of both sides and impose a constant mean to get $\mu = \phi_0/(1 - \phi_1)$; the same trick on the variance needs $\phi_1^2 < 1$. *Stationarity:* $|\phi_1| < 1$, i.e. the root of $1 - \phi_1 x = 0$ exceeds 1 in modulus; otherwise the series explodes or wanders with no fixed mean.

Markov property: only yesterday matters

$\mathbb{E}(r_t \mid r_{t-1}, r_{t-2}, \dots) = \phi_0 + \phi_1 r_{t-1}$ and $\mathrm{Var}(r_t \mid r_{t-1}) = \sigma_a^2$ – once you know yesterday, older days add nothing.

The ACF of an AR(1), step by step

Claim: the autocorrelations are $\rho_\ell = \phi_1^\ell$. Here is the reasoning, in words and one line.

Where it comes from

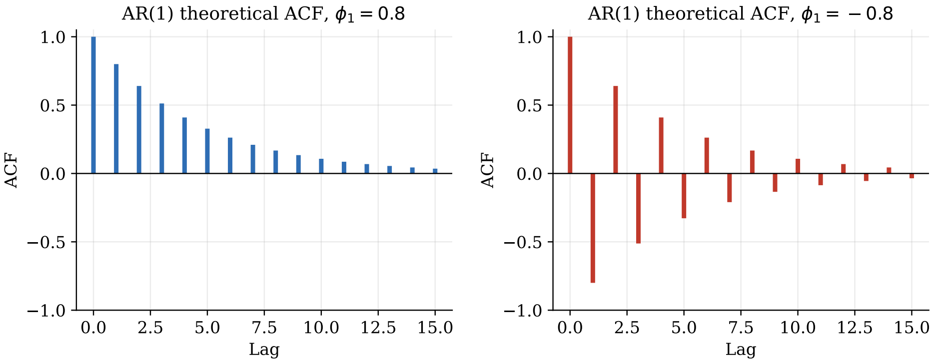
Write the model in mean-deviation form, multiply both sides by $r_{t-\ell}$, and take expectations. Because the shock a_t is uncorrelated with the past,

$$\rho_\ell = \phi_1 \, \rho_{\ell-1} \quad \Longrightarrow \quad \rho_\ell = \phi_1^\ell \quad (\text{since } \rho_0 = 1).$$

Each extra lag simply multiplies the correlation by ϕ_1, so the ACF decays **geometrically** – it tails off smoothly, never cutting off abruptly:

- ϕ_1 near $+1$: slow decay, *long memory*, trending/“momentum”.
- ϕ_1 near 0: fast decay, *short memory*.
- $\phi_1 < 0$: the sign flips at every lag – the fingerprint of *mean reversion*.

AR(1) ACF: exponential decay, two flavours



- $\rho_\ell = \phi_1^\ell$: starts at 1 and decays geometrically.
- $\phi_1 = 0.8$: persistent, smooth positive decay (trending, “momentum-like”).
- $\phi_1 = -0.8$: sign flips each lag – the signature of *mean reversion*.

AR(2): a second lag brings cycles

$$r_t = \phi_0 + \phi_1 r_{t-1} + \phi_2 r_{t-2} + a_t$$

Now today depends on the *last two* returns. That extra lag is precisely what lets the model generate **cycles**, not just smooth decay.

Stationarity

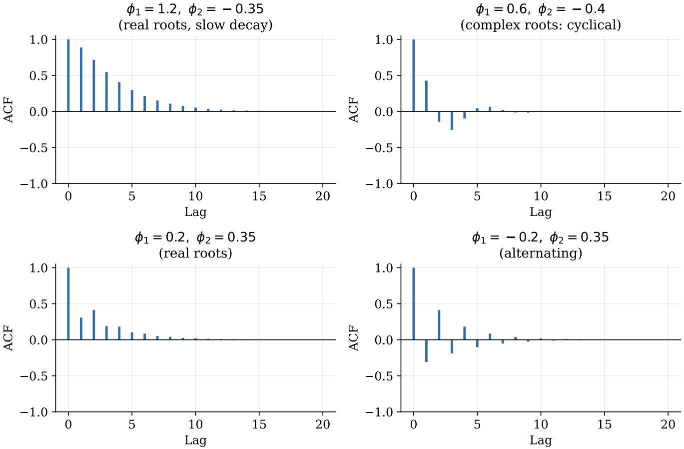
Both roots of $1 - \phi_1 x - \phi_2 x^2 = 0$ lie outside the unit circle, equivalently

$$\phi_1 + \phi_2 < 1, \qquad \phi_2 - \phi_1 < 1, \qquad |\phi_2| < 1.$$

ACF: a two-term recursion $\rho_\ell = \phi_1 \rho_{\ell-1} + \phi_2 \rho_{\ell-2}$

Real roots $\Rightarrow$ a mixture of exponential decays (possibly alternating).
Complex roots ($\phi_1^2 + 4\phi_2 < 0$) $\Rightarrow$ **damped sine waves** – a stochastic business cycle of average length $k = 2\pi \big/ \cos^{-1}\left(\frac{\phi_1}{2\sqrt{-\phi_2}}\right)$.

AR(2) ACF patterns



Top-right ($\phi_1 = 0.6, \phi_2 = -0.4$): complex roots $\Rightarrow$ oscillating, business-cycle-like ACF. The others have real roots $\Rightarrow$ pure (possibly alternating) exponential decay.

ACF vs. PACF: total link vs. direct link

Two ways to ask "how does today relate to lag ℓ?"

ACF – the *total* correlation at lag ℓ

How similar is the series to itself ℓ periods ago? It bundles together the *direct* link from $r_{t-\ell}$ *and* the *indirect* link relayed through all the days in between.

PACF – the *direct* correlation at lag ℓ

The correlation between r_t and $r_{t-\ell}$ *after* stripping out everything already explained by lags $1, \ldots, \ell - 1$. It is the genuinely new information that lag ℓ adds on its own.

Why this pins down the AR order

In an AR(p), lags beyond p have no *direct* effect – their apparent correlation is merely relayed through the first p lags. So the **PACF cuts off at lag p**, while the ACF only tails off gradually.

Identifying AR order in practice: the PACF

Idea: the lag-j **partial** autocorrelation $\phi_{j,j}$ is the last coefficient when fitting an AR(j) – the effect of r_{t-j} *after* removing lags $1, \ldots, j-1$:

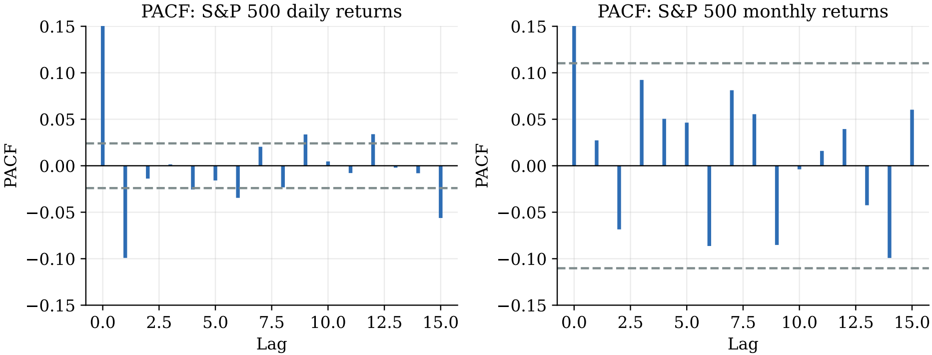
$$r_t = \phi_{0,j} + \phi_{1,j} r_{t-1} + \cdots + \phi_{j,j} r_{t-j} + e_t.$$

Key fact

For an AR(p) series, $\hat{\phi}_{j,j} \xrightarrow{p} 0$ for $j > p$ (with variance $\approx 1/T$): *the PACF cuts off at lag p.*

Model	ACF	PACF
AR(p)	tails off	**cuts off at** p

PACF on our data



- **Daily** S&P 500 (left): one significant negative spike at lag 1 $\Rightarrow$ an AR(1) is a sensible first model.
- **Monthly** (right): nothing convincingly outside the bands $\Rightarrow$ AR(0), i.e. white noise.

Order selection via information criteria

How many lags p? Two forces pull against each other:

- more lags *always* lower the in-sample residual variance $\hat{\sigma}_j^2$ – better fit;
- but every extra lag is another parameter to estimate – and risks fitting *noise*.

An information criterion = misfit + a fee per parameter

From the maximised Gaussian likelihood (divide by T, drop constants):

$$\mathrm{AIC}(j) = \ln \hat{\sigma}_j^2 + \frac{2j}{T}, \qquad \mathrm{BIC}(j) = \ln \hat{\sigma}_j^2 + \frac{j \ln T}{T}.$$

The *misfit* $\ln \hat{\sigma}_j^2$ falls with j; the *fee* (2 for AIC, $\ln T$ for BIC) rises with j.

Why a best order exists

Misfit keeps falling, the fee keeps rising: their sum dips, then climbs, so the criterion has an **interior minimum** – and that is the order we pick.

The selection rule

Rule

Fit AR(j) for $j = 0, 1, \ldots, P_{\max}$ on the *same* sample, then choose

$$\hat{p} = \arg \min_{0 \le j \le P_{\max}} \mathrm{IC}(j).$$

- State which criterion you used, and compare models on *identical* observations (same T) – otherwise the scores are not comparable.
- **BIC** is *consistent*: as $T \to \infty$ it picks the true order (heavier penalty $\ln T \gg 2$, so it leans to *smaller* models).
- **AIC** targets best out-of-sample prediction; it is not consistent and tends to choose a *slightly larger* model.

When they disagree

E.g. on *daily* S&P 500 returns BIC→AR(1) but AIC→AR(9). Prefer the **parsimonious** model unless theory or residual diagnostics demand more lags.

PACF and information criteria: monthly S&P 500 simple returns

Monthly *simple* returns $r_t = P_t/P_{t-1} - 1$, 2000–2026 ($T = 316$). Significance band $\pm 2/\sqrt{T} = \pm 0.113$; smallest IC in each column in **bold**.

p	PACF $\hat{\phi}_{p,p}$	AIC(p)	BIC(p)
0	–	**−6.255**	**−6.255**
1	0.016	−6.249	−6.237
2	−0.074	−6.248	−6.224
3	0.084	−6.249	−6.213
4	0.046	−6.245	−6.197
5	0.044	−6.241	−6.181
6	−0.080	−6.241	−6.170
7	0.079	−6.241	−6.158
8	0.057	−6.238	−6.143
9	−0.085	−6.240	−6.133
10	−0.005	−6.234	−6.115

Both criteria are smallest at $p = 0$: **AR(0)** – the series is essentially white noise.

Reading the table (1): the PACF column

What each entry is

$\hat{\phi}_{p,p}$ is the *last* coefficient when AR(p) is fitted – the direct effect of r_{t-p}. It is built recursively from the sample ACFs $\hat{\rho}_\ell$ (Durbin–Levinson):

$$\hat{\phi}_{p,p} = \frac{\hat{\rho}_p - \sum_{i=1}^{p-1} \hat{\phi}_{p-1,i}\, \hat{\rho}_{p-i}}{1 - \sum_{i=1}^{p-1} \hat{\phi}_{p-1,i}\, \hat{\rho}_i}.$$

The significance band

If the series is white noise, $\hat{\phi}_{p,p} \dot\sim N(0, 1/T)$, so anything inside $\pm 2/\sqrt{T} = \pm 0.113$ is just sampling noise.

Every entry above is inside the band (largest $|\hat{\phi}_{p,p}| = 0.085$ at lag 9): **no lag stands out**, and the PACF cuts off nowhere $\Rightarrow$ no AR order is indicated.

Reading the table (2): the AIC and BIC columns

With $T = 316$ ($\ln T = 5.76$) each row is simply

$$\mathrm{AIC}(p) = \ln \hat{\sigma}_p^2 + \frac{2p}{316}, \qquad \mathrm{BIC}(p) = \ln \hat{\sigma}_p^2 + \frac{5.76\,p}{316}.$$

Worked row: does lag 1 earn its keep?

Going $p = 0 \to 1$ the fit barely moves: $\ln \hat{\sigma}_0^2 = -6.2550$ vs $\ln \hat{\sigma}_1^2 = -6.2553$ (a gain of 0.0003). But AIC adds $2/316 = 0.0063$ and BIC adds $5.76/316 = 0.018$. *Fee* $>$ *gain*, so both scores **rise**.

Verdict

No extra lag pays for itself; both criteria bottom out at $p = 0$. Monthly S&P 500 returns are **white noise** – exactly what the PACF said, and what efficient markets predict at the monthly horizon.

Parameter estimation

Given the order p, estimate $(\phi_0, \ldots, \phi_p, \sigma_a^2)$. Three routes, equivalent in large samples:

Three estimators

1. **Conditional least squares** – treat the first p values as fixed and run OLS of r_t on its p lags. Simplest; what we use.
2. **Yule–Walker** – plug sample ACFs into $\rho_\ell = \phi_1 \rho_{\ell-1} + \cdots + \phi_p \rho_{\ell-p}$ and solve.
3. **Exact maximum likelihood** (Gaussian) – uses all T observations. R: `arima(x, order=c(p,0,0), method="ML")`.

For AR(1), OLS gives $\hat{\phi}_1 = \widehat{\text{Cov}}(r_t, r_{t-1}) / \widehat{\text{Var}}(r_{t-1}) = \hat{\rho}_1$ and $\hat{\phi}_0 = \bar{r}\,(1 - \hat{\phi}_1)$.

Daily S&P 500 AR(1) ($T = 6{,}640$)

$\hat{\phi}_1 = -0.099$ (s.e. 0.012, $t = -8.1$): strongly significant. $\hat{\phi}_0 = 2.7 \times 10^{-4}$, $\hat{\sigma}_a = 0.0121$, implied $\hat{\mu} = 2.5 \times 10^{-4}$.

Model checking: are the residuals white noise?

A fitted AR is adequate only if its residuals $\hat{a}_t$ carry *no* leftover structure.

Three checks on $\hat{a}_t$

1. **Time plot** – no trend or drift by eye.
2. **Residual ACF** – all spikes inside $\pm 2/\sqrt{T}$.
3. **Ljung–Box** $Q(m) = T(T+2)\sum_{\ell=1}^{m} \dfrac{\hat{\rho}_\ell^2}{T-\ell}$, compared with χ^2_{m-p} (p d.f. lost to estimation). Large Q (small p-value) $\Rightarrow$ structure remains.

Daily AR(1): a useful but imperfect first model

$Q(10) = 28.7$ on $10 - 1 = 9$ d.f. ($p = 0.001$): residuals are *not* fully white. The AR(1) captures the lag-1 effect but misses weak higher-order dependence – a cue to consider more lags (or an ARMA/GARCH refinement later).

Goodness of fit

R^2: the fraction of variance explained

$$R^2 = 1 - \frac{\sum_t \hat{a}_t^2}{\sum_t (r_t - \bar{r})^2} = 1 - \frac{\hat{\sigma}_a^2}{\hat{\sigma}_r^2}.$$

Adding lags can only *raise* in-sample R^2, so to *compare* orders use the adjusted R^2 (or AIC/BIC), which charge for extra parameters.

For returns, R^2 is tiny by design

Daily S&P 500 AR(1): $R^2 \approx 0.010$ – the model explains only $\sim$1% of return variance. That is *expected*: strong predictability would be arbitraged away.

The right target

A high R^2 is *not* the goal – **white-noise residuals** are. A model can have $R^2 \approx 0$ and still be "correct" if it leaves no exploitable structure.

Forecasting with an AR model: the goal

Standing at the **origin** h with data up to r_h, predict $r_{h+\ell}$ for horizon $\ell \geq 1$.

The optimal point forecast

The forecast that minimises mean-squared error is the **conditional expectation**

$$\hat{r}_h(\ell) = \mathbb{E}(r_{h+\ell} \mid r_h, r_{h-1}, \dots) \,.$$

Recipe for AR models: *write the equation for* $r_{h+\ell}$*, then replace each unknown future return by its forecast and each future shock* a *by* 0 (its mean).

- Forecast error: $e_h(\ell) = r_{h+\ell} - \hat{r}_h(\ell)$; its variance gives the interval.
- We build $\ell = 1$, then $\ell = 2$, then general ℓ – each from the previous one.

1-step-ahead forecast

Write the model one step past the origin:

$$r_{h+1} = \phi_0 + \phi_1 r_h + \cdots + \phi_p r_{h+1-p} + a_{h+1}.$$

Everything on the right is known except the future shock a_{h+1}; put $\mathbb{E}(a_{h+1}) = 0$:

Forecast and error

$$\hat{r}_h(1) = \phi_0 + \phi_1 r_h + \cdots + \phi_p r_{h+1-p}, \qquad e_h(1) = a_{h+1}.$$

The 1-step error *is* the next shock, so $\mathrm{Var}[e_h(1)] = \sigma_a^2$ – the smallest possible.

Daily AR(1), origin $r_h = 0.0022$

$\hat{r}_h(1) = \hat{\mu} + \hat{\phi}_1(r_h - \hat{\mu}) = 5.8 \times 10^{-5}$ (essentially μ); 95% band $\approx \hat{r}_h(1) \pm 2\hat{\sigma}_a = \pm 0.024$.

2-step-ahead forecast

Now write the model two steps out (AR(1) for clarity):

$$r_{h+2} = \phi_0 + \phi_1 r_{h+1} + a_{h+2}.$$

r_{h+1} is unknown, so replace it by its forecast $\hat{r}_h(1)$, and set future shocks to 0:

Forecast and error

$$\hat{r}_h(2) = \phi_0 + \phi_1 \hat{r}_h(1), \qquad e_h(2) = a_{h+2} + \phi_1 a_{h+1}.$$

Two shocks now contribute: $\mathrm{Var}[e_h(2)] = \sigma_a^2(1+\phi_1^2) \geq \sigma_a^2$.

Daily AR(1)

$\hat{r}_h(2) = \hat{\mu} + \hat{\phi}_1^2(r_h - \hat{\mu}) = 2.7 \times 10^{-4}$; the band widens to $\pm 2\hat{\sigma}_a\sqrt{1+\hat{\phi}_1^2}$. **Uncertainty grows with the horizon.**

Multi-step-ahead forecast

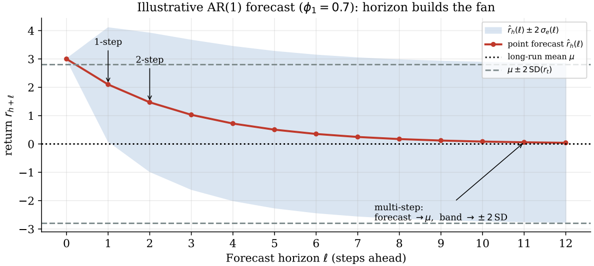
Iterate the same idea – forecast forward, plug forecasts back in:

$$\hat{r}_h(\ell) = \phi_0 + \phi_1 \hat{r}_h(\ell - 1) + \cdots + \phi_p \hat{r}_h(\ell - p),$$

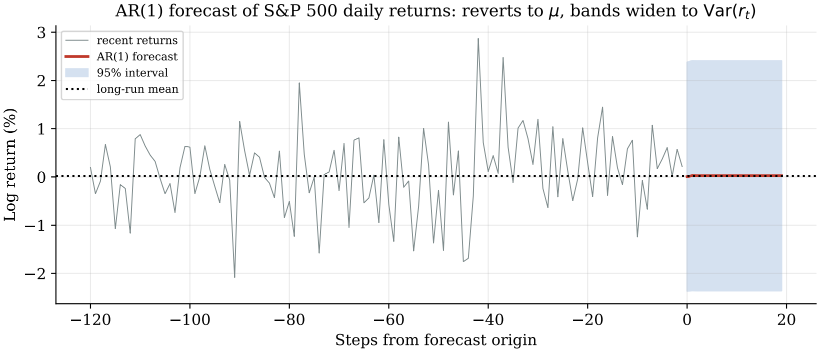
with error $e_h(\ell) = \sum_{j=0}^{\ell-1} \psi_j \, a_{h+\ell-j}$ and variance $\sigma_e^2(\ell) = \sigma_a^2 \sum_{j=0}^{\ell-1} \psi_j^2$ (for AR(1), $\psi_j = \phi_1^j$).

Two limits as $\ell \to \infty$ (stationary AR)

$\hat{r}_h(\ell) \to \mu$ (forecasts **mean-revert**) and $\sigma_e^2(\ell) \to \sigma_a^2/(1 - \phi_1^2) = \mathrm{Var}(r_t)$ (bands stabilise at $\pm 2\,$SD).



Forecasting on real data



- AR(1) on S&P 500 *daily* returns: because $\hat{\phi}_1 = -0.099$ is small, the point forecast reverts to μ **within a couple of days**.
- The 95% band almost instantly reaches $\pm 2\,\mathrm{SD}(r_t)$: beyond a day or two, the best guess is just the unconditional mean.
- Takeaway: AR models forecast the *near* horizon – and for equities that horizon is very short.

Moving-Average (MA) Models

Moving-average models: two ways in

An MA model expresses today's return through *past shocks* a_{t-j} rather than past returns r_{t-j}.

Two routes to the same model

1. *As an extension of white noise* – let r_t be a short, weighted sum of the current and a few recent shocks $a_t, a_{t-1}, \ldots$
2. *As an infinite-order AR with constraints* – start from an AR(∞) and tie its infinitely many coefficients to just a *handful* of parameters.

We adopt route 2

It shows *where* MA models come from and *why* the invertibility condition later appears – the AR(∞) viewpoint makes both feel natural.

From an AR(∞) to an MA(1), step by step

1. **Start with an AR of infinite order** – exact, but impossible to estimate:

$$r_t = \phi_0 + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \cdots + a_t.$$

2. **Constrain the coefficients** to one parameter, $\phi_i = -\theta_1^i$ (need $|\theta_1| < 1$ so the sum converges):

$$r_t = \phi_0 - \theta_1 r_{t-1} - \theta_1^2 r_{t-2} - \cdots + a_t \iff \sum_{i \geq 0} \theta_1^i r_{t-i} = \phi_0 + a_t.$$

3. **Subtract $\theta_1 \times$(the same equation at $t-1$).** Since $\sum_{i\geq 0} \theta_1^i r_{t-i} - \theta_1 \sum_{i \geq 0} \theta_1^i r_{t-1-i} = r_t$, the infinite sum telescopes to

$$r_t = \phi_0(1-\theta_1) + a_t - \theta_1 a_{t-1} = c_0 + a_t - \theta_1 a_{t-1}.$$

That is an MA(1)

Infinite memory in past *returns* collapses to *finite* memory in past *shocks*. Keeping q shock terms gives the general MA(q).

Properties: MA(1) and MA(2)

$$\text{MA}(q): \quad r_t = c_0 + a_t - \theta_1 a_{t-1} - \cdots - \theta_q a_{t-q}, \qquad \{a_t\} \sim \text{WN}(0, \sigma_a^2).$$

MA(1): $r_t = c_0 + a_t - \theta_1 a_{t-1}$

$$\mathbb{E}(r_t) = c_0, \quad \text{Var}(r_t) = (1+\theta_1^2)\sigma_a^2, \quad \rho_1 = \frac{-\theta_1}{1+\theta_1^2}, \quad \rho_\ell = 0 \ (\ell > 1).$$

MA(2): $r_t = c_0 + a_t - \theta_1 a_{t-1} - \theta_2 a_{t-2}$

$\text{Var}(r_t) = (1+\theta_1^2+\theta_2^2)\sigma_a^2$, and

$$\rho_1 = \frac{-\theta_1(1-\theta_2)}{1+\theta_1^2+\theta_2^2}, \qquad \rho_2 = \frac{-\theta_2}{1+\theta_1^2+\theta_2^2}, \qquad \rho_\ell = 0 \ (\ell > 2).$$

The mean is free of the θ's, and the **ACF cuts off after lag** q.

MA models and stationarity

MA(q) is *always* weakly stationary

r_t is a *finite* weighted sum of white-noise terms, so for *any* $\theta_1, \ldots, \theta_q$:

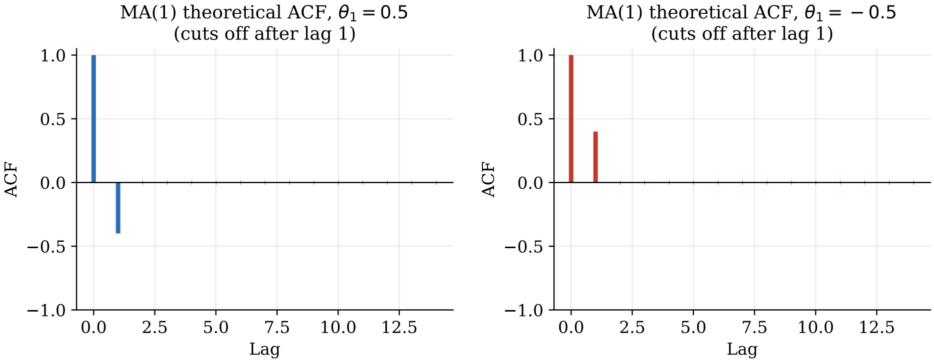
$$\mathbb{E}(r_t) = c_0 \text{ (constant)}, \qquad \mathrm{Var}(r_t) = (1 + \theta_1^2 + \cdots + \theta_q^2)\sigma_a^2 \text{ (finite, constant)},$$

and every autocovariance depends on the lag alone. *No condition on the θ's is needed.*

Contrast with AR

AR models need $|\phi|$ restrictions to be stationary; MA models get stationarity **for free**. The price MA pays instead is the **invertibility** condition – next.

MA(1) ACF: one spike, then silence



- $\rho_1 = -\theta_1/(1+\theta_1^2)$ and $\rho_\ell = 0$ for every $\ell > 1$: the ACF **cuts off after lag 1**.
- The sign of ρ_1 is *opposite* to the sign of θ_1.
- Hard ceiling: $|\rho_1| \le 0.5$ for any MA(1) – the lag-1 correlation can never be large.

Invertibility, in plain English

The puzzle

θ_1 and $1/\theta_1$ produce the *same* ρ_1 – two different models, one ACF. Which one is "the" model?

The fix: keep the version we can invert

Invertibility means we can turn the MA back into a sensible AR(∞) – writing today's *shock* as a *decaying* function of observed past returns:

$$a_t = (r_t - c_0) + \theta_1 a_{t-1} = \cdots = \sum_{i \geq 0} \theta_1^i \, (r_{t-i} - c_0).$$

This sum converges only when $|\theta_1| < 1$ – so we always choose that root.

Why it matters

If $|\theta_1| > 1$ the weights *explode*: the distant past would matter *more* than the recent past – nonsense for forecasting. Software picks the invertible root for you.

Identifying the MA order q

The rule: read the sample ACF

For an MA(q) the ACF is *zero beyond lag q*. So take q as the last lag whose sample $\hat{\rho}_\ell$ pokes outside the $\pm 2/\sqrt{T}$ band; deeper spikes are noise.

The AR/MA mirror

Model	ACF	PACF
AR(p)	tails off	**cuts off at p**
MA(q)	**cuts off at q**	tails off

MA identification is the exact dual of AR identification.

Estimation: MA needs likelihood, not OLS

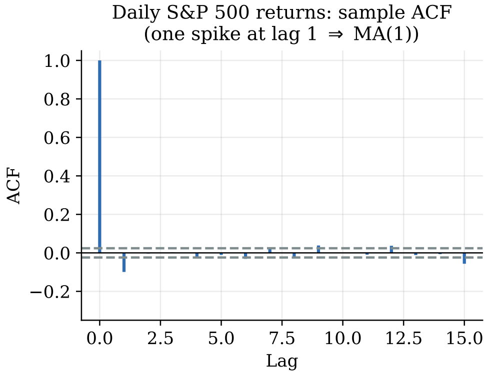
Why not OLS? The regressors $a_{t-1}, \ldots, a_{t-q}$ are *unobserved* shocks. Use **conditional** or **exact maximum likelihood**, which reconstruct the a_t recursively from the data. R: `arima(x, order=c(0,0,q))`.

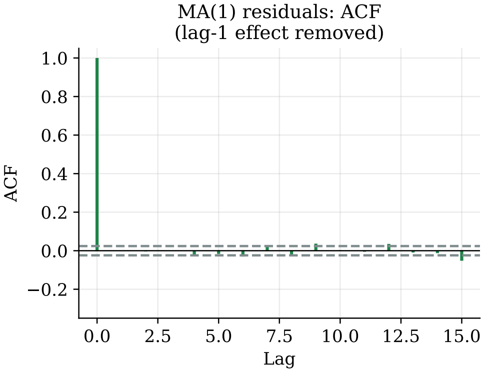
Daily S&P 500 returns: AR(1) vs. MA(1) ($T = 6{,}640$)

Model	key coefficient	s.e.	$\hat{\sigma}_a$	AIC
AR(1)	$\hat{\phi}_1 = -0.099$	0.012	0.01211	−39,769
MA(1)	$\hat{\theta}_1 = \quad 0.101$	0.012	0.01210	−39,771

The two fit *equally well*: the same lag-1 *bid–ask bounce* (§2.2) reads as AR(1) *or* MA(1). Implied $\rho_1 = -\theta_1/(1+\theta_1^2) = -0.10$, matching the data.

Estimation, checked: did the MA(1) clean the ACF?





- **Left:** daily returns show a single ACF spike at lag 1 – the MA(1) signature.
- **Right:** after fitting MA(1) the lag-1 spike is gone and residuals are nearly white (a few tiny higher-lag spikes remain, $Q(10) = 28$, $p = 0.001$ – the same mild imperfection as the AR(1)).

Forecasting with an MA model

Replace future shocks by 0 and known shocks by their fitted values. For MA(q):

$$\hat{r}_h(\ell) = \begin{cases} c_0 - \theta_\ell a_h - \cdots - \theta_q a_{h+\ell-q}, & \ell \leq q, \\ c_0, & \ell > q. \end{cases}$$

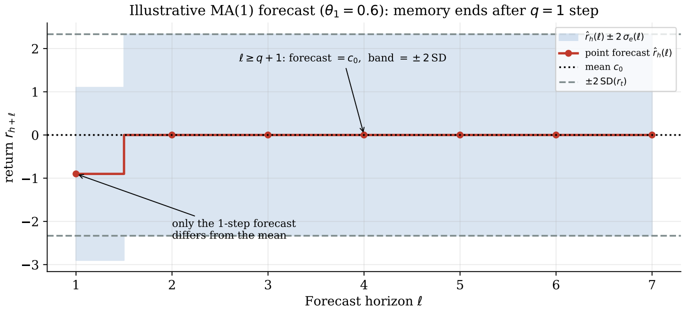
MA(1) made explicit

1-step: $\hat{r}_h(1) = c_0 - \theta_1 a_h$, error a_{h+1}, variance σ_a^2.
$\ell \geq 2$: $\hat{r}_h(\ell) = c_0$ (the mean), variance $(1 + \theta_1^2)\sigma_a^2 = \mathrm{Var}(r_t)$.

Finite memory $\Rightarrow$ an abrupt cutoff

An MA(q) "forgets" everything older than q periods: the forecast hits the mean *exactly* at horizon $q + 1$ – contrast the AR's *gradual* geometric fade.

MA forecast: a step, not a fade



Illustrative MA(1): only the 1-step forecast leaves the mean; from $\ell = 2$ on the point forecast is exactly c_0 and the band locks at $\pm 2\,\mathrm{SD}(r_t)$ – the hallmark of finite memory.

ARMA Models

ARMA: combine the two kinds of memory

An AR remembers past *returns*; an MA remembers past *shocks*. An ARMA uses both:

$$\text{ARMA}(1,1): \quad r_t = \phi_0 + \underbrace{\phi_1 r_{t-1}}_{\text{AR: echo of } r} + a_t \underbrace{- \theta_1 a_{t-1}}_{\text{MA: echo of } a}.$$

Conditions

Stationarity $|\phi_1| < 1$ (the AR part); *invertibility* $|\theta_1| < 1$ (the MA part); and $\phi_1 \neq \theta_1$ – otherwise the two terms cancel and nothing is left.

ACF/PACF signature

Because the model blends both memories, **both the ACF and the PACF tail off** (neither cuts off). That "both decay" pattern is the fingerprint that says *mixed* model.

Where ARMA comes from, in plain terms

The build story

Fit an AR and look at what is left over. If a *single-period shock echo* still sits in the residuals, don't keep piling on AR lags – add *one* MA term to mop it up. The result, an AR with a little MA attached, is an **ARMA** model.

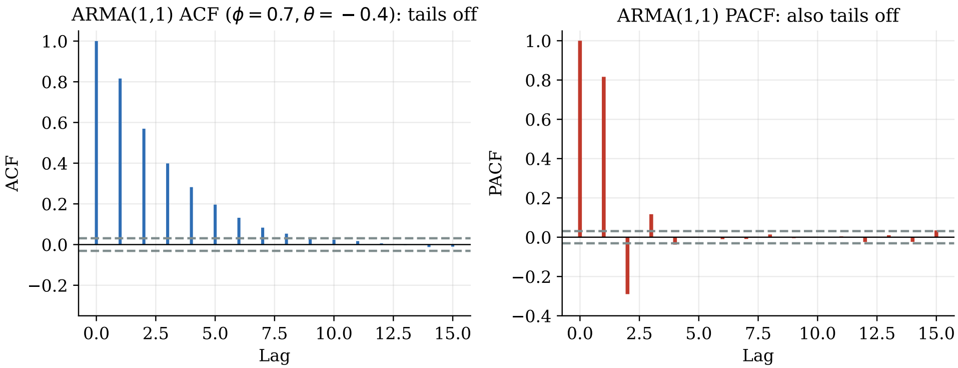
General ARMA(p,q) in backshift form $(\mathcal{B}r_t = r_{t-1})$:

$$\underbrace{\left(1 - \phi_1\mathcal{B} - \cdots - \phi_p\mathcal{B}^p\right)}_{\phi(\mathcal{B})} r_t = \phi_0 + \underbrace{\left(1 - \theta_1\mathcal{B} - \cdots - \theta_q\mathcal{B}^q\right)}_{\theta(\mathcal{B})} a_t.$$

Why bother?

One small ARMA(1,1) can stand in for a *high*-order pure AR or MA – fewer parameters, less estimation noise (**parsimony**). The same algebra returns for *volatility*: GARCH(p,q) is ARMA in a_t^2 (Ch. 3).

ARMA(1,1): both ACF and PACF tail off



Simulated ARMA(1,1), $\phi = 0.7$, $\theta = -0.4$. Neither function cuts off cleanly – this "both decay" pattern is the fingerprint that says *mixed* model.

Identifying ARMA order: the Extended ACF (EACF)

Plain ACF/PACF cannot separate p from q (both tail off). The **EACF** finds (p, q) *jointly*: it filters out the AR part, leaving a pure MA whose ACF *does* cut off.

How to read it

Each cell is **O** (autocorrelation insignificant) or **X** (significant). Find the *upper-left vertex of the triangle of O's* – its coordinates are the chosen (p, q).

	MA order q						
AR p	0	1	2	3	4	5	6
0	X	X	X	X	X	X	X
1	X	O	O	O	O	O	O
2	X	X	O	O	O	O	O
3	X	X	X	O	O	O	O

Vertex at row 1, col 1 $\Rightarrow$ **ARMA(1,1)**. In practice combine the EACF with AIC/BIC and residual checks. R: `TSA::eacf(x)`, `forecast::auto.arima`.

One ARMA model, three representations

Three equivalent ways to write the same model

1. **ARMA form:** $\phi(\mathcal{B})r_t = \phi_0 + \theta(\mathcal{B})a_t$ – compact, used for estimation.
2. **MA(∞):** $r_t = \mu + \sum_{i\geq 0} \psi_i a_{t-i}$, with $\psi(\mathcal{B}) = \theta(\mathcal{B})/\phi(\mathcal{B})$ – gives the variance, ACF and forecast-error variance (the ψ-weights).
3. **AR(∞):** $\pi(\mathcal{B})r_t = \text{const} + a_t$, with $\pi(\mathcal{B}) = \phi(\mathcal{B})/\theta(\mathcal{B})$ – shows how forecasts weight the observed past; needs invertibility.

One model, three lenses – pick whichever makes the quantity you want easy.

Forecasting an ARMA model: the equations

Same recipe as before – at origin h, replace future returns by forecasts, future shocks by 0, and past shocks by the fitted residuals.

Point forecast (ARMA(1,1))

$$\hat{r}_h(1) = \phi_0 + \phi_1 r_h - \theta_1 a_h, \qquad \hat{r}_h(\ell) = \phi_0 + \phi_1 \hat{r}_h(\ell - 1) \quad (\ell \geq 2).$$

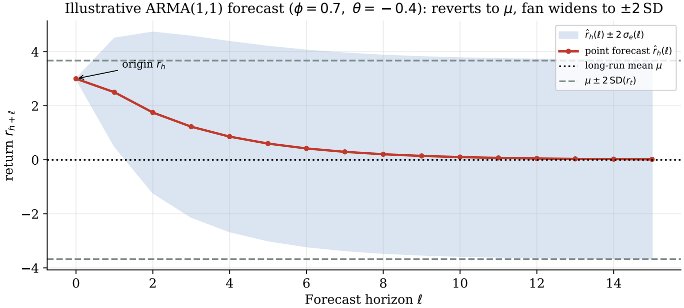
The MA term bites *once* (at $\ell = 1$); from then on the AR recursion takes over.

Error variance via the MA(∞) weights

With $\psi_0 = 1,\ \psi_1 = \phi_1 - \theta_1,\ \psi_j = \phi_1 \psi_{j-1}$,

$$\sigma_e^2(\ell) = \sigma_a^2 \sum_{j=0}^{\ell-1} \psi_j^2 \quad \xrightarrow{\ell \to \infty} \quad \mathrm{Var}(r_t), \qquad \hat{r}_h(\ell) \to \mu.$$

Forecasting an ARMA model: the picture



- The MA term nudges only the *first* step; thereafter the point forecast fades **geometrically** to μ at rate ϕ_1 – just like an AR(1).
- The fan widens with the ψ-weights and settles at $\pm 2\,\mathrm{SD}(r_t)$.
- *Same shape as the AR fan*: blending AR and MA buys parsimony, not a new forecast profile.

Unit-Root Nonstationarity

Unit roots: the random walk

A **random walk** is an AR(1) sitting exactly on the stationarity boundary, $\phi_1 = 1$:

$$p_t = p_{t-1} + a_t.$$

Why it is non-stationary

- Variance grows without bound: $\mathrm{Var}(p_t \mid p_0) = t\,\sigma_a^2$.
- Shocks are *permanent* – each a_t stays in the level forever; $\rho_\ell \to 1$ for any fixed lag.
- The best forecast of *every* future value is simply today's value p_t.

Where "unit root" comes from

The AR polynomial $1 - \phi_1 \mathcal{B} = 1 - \mathcal{B}$ has its root at $\mathcal{B} = 1$ – *on* the unit circle, not outside it. That single root sitting on the circle is exactly what breaks stationarity.

Random walk with drift

Add a constant μ to the random walk:

$$p_t = \mu + p_{t-1} + a_t.$$

What the constant does

Summing from the start, $p_t = p_0 + \mu\, t + \sum_{i=1}^{t} a_i$. So the constant μ is a **drift**: an *average change per period* that cumulates into a *deterministic linear trend* μt, laid on top of the random-walk wandering $\sum a_i$.

Reading the drift

For log prices, $\mu = \mathbb{E}(p_t - p_{t-1}) = \mathbb{E}(r_t)$ is the *mean return* – the slope of the long-run price path. A positive μ is why equity prices trend upward over decades.

Two ingredients now: a deterministic slope μt *and* a stochastic trend $\sum a_i$.

Trend-stationary vs. difference-stationary

A **trend-stationary** series has a *deterministic* trend with stationary noise around it:

$$p_t = \beta_0 + \beta_1 t + e_t, \qquad e_t \text{ stationary.}$$

The crucial difference

- *Trend-stationary:* shocks are **temporary** – the series always returns to the line $\beta_0 + \beta_1 t$. Cure: *detrend* (regress on t, keep the residual).
- *Difference-stationary* (unit root): shocks are **permanent** – there is no fixed line to return to. Cure: *difference* (Δp_t).

Why you must not confuse them

Both can look like an upward-sloping path. Detrending a true unit root – or differencing a true trend-stationary series – gives *wrong* standard errors and spurious forecasts.

Unit-root models and differencing

The cure: difference until stationary

If p_t has a unit root, its first difference is stationary:

$$\Delta p_t = (1 - \mathcal{B})p_t = p_t - p_{t-1}.$$

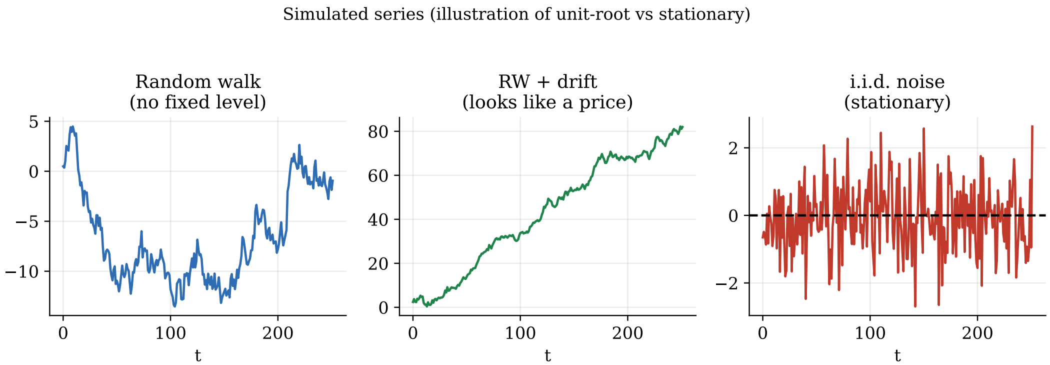
A series needing d differences to become stationary is **integrated of order d**, $I(d)$; modelling $\Delta^d p_t$ as ARMA(p, q) gives an **ARIMA(p, d, q)**.

The core finance fact

Stock *log prices* are $I(1)$ (unit-root non-stationary); log *returns* $r_t = \ln P_t - \ln P_{t-1} = \Delta \ln P_t$ are $I(0)$ (stationary). *One* difference does it – which is precisely why we model returns, not prices.

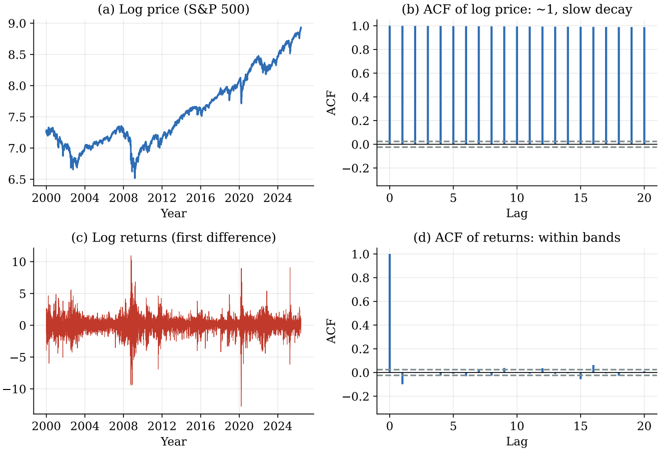
Caution: do not *over*-difference – differencing an already-stationary series injects an artificial MA unit root.

Difference vs. deterministic trend (simulated)



A random walk with drift (centre) mimics a price path – but the right cure is *differencing*, not detrending. Mistaking one for the other gives wrong standard errors and spurious forecasts.

Price vs. return on real data



(b) ACF of the log *price* stays near 1 for many lags ⇒ unit root. (d) ACF of the log *return* sits inside the bands ⇒ stationary. One difference turns the left column into the right.

Unit-root tests: the Augmented Dickey–Fuller (ADF) test

Subtract p_{t-1} from an AR and test whether the level term vanishes (constant case):

$$\Delta p_t = \alpha_0 + \beta\, p_{t-1} + \sum_{i=1}^{k} \gamma_i\, \Delta p_{t-i} + e_t, \qquad H_0 : \beta = 0 \text{ (unit root)} \;\; \text{vs.} \;\; H_a : \beta < 0 \text{ (stationary)}.$$

A *more negative* statistic $\Rightarrow$ reject the unit root. The $\sum \gamma_i \Delta p_{t-i}$ terms (the "augmentation") soak up short-run autocorrelation so the test stays valid.

Non-standard distribution

Under H_0 the statistic is *not* $\mathcal{N}(0,1)$ or Student-*t* – use Dickey–Fuller critical values (e.g. -2.86 at 5%, constant case).

S&P 500 (daily, our data)

Log *price*: ADF $= +1.06 \Rightarrow$ *fail to reject* (unit root). Log *return*: ADF $= -27.7 \Rightarrow$ *reject* decisively (stationary).

Choose lag k by AIC/BIC or $k \approx (4T/100)^{1/4}$. R: `urca::ur.df(x, type="drift", selectlags="AIC")`; `tseries::adf.test`.

Seasonal Models

Seasonality and seasonal differencing

Many series repeat every s periods ($s = 12$ monthly, $s = 4$ quarterly): earnings, dividends, macro data, retail sales.

Seasonal differencing

$$\Delta_s r_t = r_t - r_{t-s} = (1 - \mathcal{B}^s) r_t,$$

often combined with a regular difference, $(1 - \mathcal{B})(1 - \mathcal{B}^{12}) r_t$, to remove trend *and* season.

Multiplicative SARIMA $(p, d, q) \times (P, D, Q)_s$

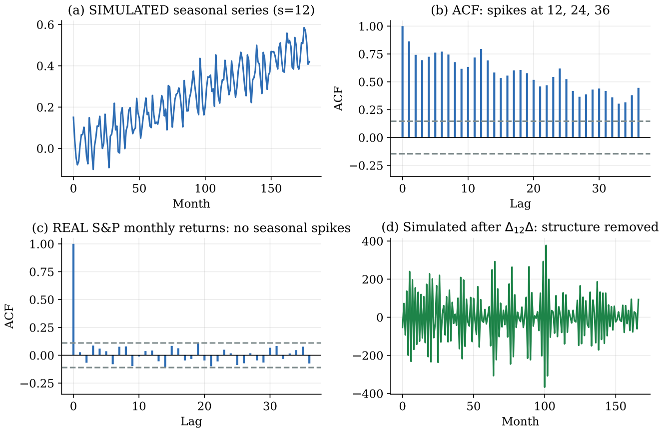
$$\Phi(\mathcal{B}^s)\, \phi(\mathcal{B})\, (1 - \mathcal{B})^d (1 - \mathcal{B}^s)^D r_t = \Theta(\mathcal{B}^s)\, \theta(\mathcal{B})\, a_t.$$

Regular polynomials ϕ, θ act in $\mathcal{B}$; seasonal polynomials Φ, Θ act in $\mathcal{B}^s$.

An honest check on our data

Equity *returns* are *not* seasonal: the monthly S&P 500 ACF at lags $12, 24, 36$ is ≈ 0.04, 0.02, -0.07 – all insignificant. So we teach the method on a clearly-labelled simulated series.

Seasonality: method vs. reality



(a,b) *Simulated* s=12 series: ACF spikes at 12/24/36. (d) After $\Delta_{12}\Delta$ the seasonal structure is gone. (c) *Real* monthly S&P returns: no seasonal spikes – markets price the calendar away.

Regression Models with Time Series Errors

Regression when the errors are serially correlated

Model $y_t = \alpha + \beta x_t + e_t$ with $\{e_t\}$ autocorrelated (the norm for financial levels, e.g. two price or yield series).

What breaks, and what doesn't

- $\hat{\beta}_{\text{OLS}}$ stays *consistent* (under mild conditions).
- Its *standard errors are wrong* $\Rightarrow$ t- and F-tests are invalid, usually *overstating* significance.

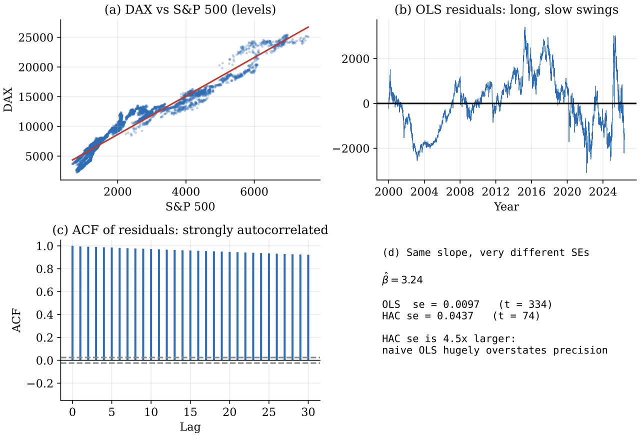
(a) Model the error

Fit ARMA errors jointly (GLS):
$y_t = \alpha + \beta x_t + u_t$, $\phi(\mathcal{B})u_t = \theta(\mathcal{B})a_t$.
R: `arima(y, xreg=x, order=...)`

(b) Fix the SEs

Keep OLS $\hat{\beta}$, use HAC (Newey–West) standard errors – §2.10.

A concrete warning: DAX on S&P 500



Residuals from regressing the DAX *level* on the S&P 500 *level* are hugely autocorrelated (lag-1 ACF ≈ 0.995). The slope is unchanged, but the **HAC standard error is $4.5\times$ the naive OLS one** – the OLS t of 334 is a fiction; the honest t is 74. (Both series are near unit-root: differencing or cointegration analysis is the deeper fix.)

Consistent Covariance Matrix Estimation

HAC / Newey–West covariance

When errors are heteroskedastic *and* autocorrelated, estimate the "meat" of the sandwich robustly:

$$\widehat{\mathrm{Var}}(\hat{\beta})_{\mathrm{HAC}} = (X^\top X)^{-1}\,\hat{\Omega}\,(X^\top X)^{-1},$$

$$\hat{\Omega} = \hat{\Gamma}_0 + \sum_{j=1}^{L} w_j\big(\hat{\Gamma}_j + \hat{\Gamma}_j^\top\big), \qquad w_j = 1 - \frac{j}{L+1}\ \text{(Bartlett)}.$$

Why the weights?

The Bartlett weights w_j taper distant autocovariances so $\hat{\Omega}$ is positive semi-definite. Bandwidth L grows slowly with T (e.g. $L \approx 4(T/100)^{2/9}$).

Use it when

You trust the *coefficients* but not the *i.i.d.* assumption – the standard "robust SE" move in empirical finance.

R: `lmtest::coeftest(fit, vcov=sandwich::NeweyWest(fit))`

Long-Memory Models

Long memory: between stationary and unit root

Fractional differencing

$$(1-\mathcal{B})^d x_t = a_t, \qquad -0.5 < d < 0.5$$

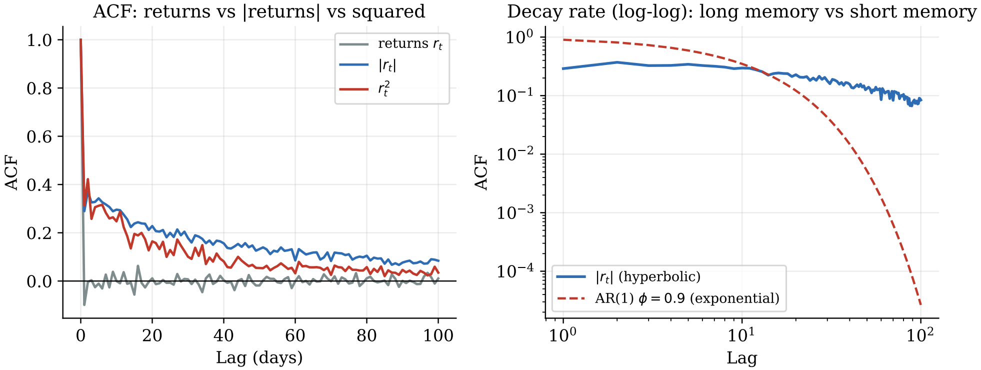
- $0 < d < 0.5$: **long memory** – ACF decays *hyperbolically*, $\rho_\ell \sim \ell^{2d-1}$ (not summable), yet the series is still stationary and mean-reverting.
- $d = 0$: white noise; $d = 1$: random walk; $-0.5 < d < 0$: anti-persistent.

ARFIMA(p, d, q)

$\phi(\mathcal{B})(1-\mathcal{B})^d r_t = \phi_0 + \theta(\mathcal{B}) a_t$ – short-run ARMA dynamics plus a long-run memory parameter d.

R: `fracdiff::fracdiff(x, nar, nma)`.

Volatility remembers; returns forget



S&P 500 daily data. *Returns* r_t (grey) are serially uncorrelated. But $|r_t|$ and r_t^2 stay positively autocorrelated for *hundreds* of lags – estimated $d \approx 0.43$ for $|r_t|$. The log–log panel shows the tell-tale straight-line (hyperbolic) decay vs. the steep exponential fall of an AR(1). *This long memory in volatility is the empirical springboard for Chapter 3.*

Chapter 2 in one slide

Concepts

- Weak stationarity = constant mean + lag-only autocovariance.
- ACF & PACF classify the model; Ljung–Box tests independence.

Models

- AR(p): PACF cuts off at p.
- MA(q): ACF cuts off at q.
- ARMA(p, q): both tail off.
- ARFIMA: hyperbolic (long) memory.

Building & checking

- AIC/BIC and EACF for order; residual ACF + $Q(m)$ for adequacy.
- Forecasts revert to μ; bands widen to $\mathrm{Var}(r_t)$.

Non-stationarity

- Prices have a unit root; returns are stationary (ADF).
- Serially correlated regression errors $\Rightarrow$ HAC SEs.

The empirical thread of this lecture

Daily index returns: weak (negative) AR(1). Monthly returns: white noise. Prices: unit root. Volatility: long memory. *Next: model that volatility – ARCH/GARCH.*

Check your understanding

1. A series has an ACF that cuts off after lag 2 and a PACF that tails off. Which model?
2. Why can a random walk's sample ACF look "significant" at long lags even though the series is not stationary?
3. You regress one yield on another and get a t-statistic of 300. Why should you be suspicious before celebrating?
4. Returns are uncorrelated but |returns| are strongly autocorrelated. What does this rule out, and what does it motivate?
5. BIC picks AR(1); AIC picks AR(7). Which do you report for a forecasting model, and why?

(Answers: MA(2); spurious persistence under a unit root; serially correlated errors invalidate OLS SEs; rules out i.i.d., motivates GARCH; BIC – parsimony and consistency.)

Appendix

Reproducible R code on the local `data/` folder

Full walk-through in `FI362_Ch2_Supplement.qmd`.

R: load local data & sample ACF / Ljung–Box

```
data_dir <- "../data"                                  # course data folder (no downloads)
idx <- read.csv(file.path(data_dir, "indices_adjclose.csv"), check.names = FALSE)
spm <- read.csv(file.path(data_dir, "sp500_monthly.csv"))

logret <- function(p) diff(log(p[is.finite(p)]))
sp_d <- logret(idx$SP500)                              # daily S&P 500 log returns
sp_m <- logret(spm$SP500)                              # monthly S&P 500 log returns

acf(sp_d, lag.max = 30)                                 # negative lag-1 spike (~ -0.10)
Box.test(sp_d, lag = 10, type = "Ljung-Box") # p < 1e-15  -> dependence
Box.test(sp_m, lag = 12, type = "Ljung-Box") # p ~ 0.20    -> white noise
```

R: AR/ARMA selection, ADF, HAC, long memory

```
library(forecast); library(urca); library(sandwich); library(lmtest)

## Order selection + diagnostics (daily S&P 500 returns)
fit <- auto.arima(sp_d, max.p = 7, max.q = 7, stationary = TRUE,
                  seasonal = FALSE, ic = "bic", stepwise = FALSE)
checkresiduals(fit)                              # BIC -> AR(1), phi1 ~ -0.099

## Unit-root test: price has one, return does not
lp <- log(idx$SP500[is.finite(idx$SP500)])
summary(ur.df(lp,   type = "drift", selectlags = "AIC"))  # fail to reject
summary(ur.df(sp_d, type = "none",  selectlags = "AIC"))  # reject

## Regression with serially correlated errors: DAX on S&P (levels)
df  <- na.omit(idx[, c("SP500","DAX")])
reg <- lm(DAX ~ SP500, data = df)
coeftest(reg, vcov = NeweyWest(reg, lag = 20)) # HAC se ~ 4.5x OLS se

## Long memory in volatility
library(fracdiff)
fracdiff(abs(sp_d), nar = 0, nma = 0)$d        # d ~ 0.43
```

Thank you

Questions?

Reproduce every number in this deck: `python make_figures.py` or render `FI362_Ch2_Supplement.qmd` in R.